

POWER FACTOR CORRECTION OF INDUCTION MOTORS USING PWM INVERTER FED AUXILIARY STATOR WINDING

Indraman Tamrakar
Institute of Engineering, Tribhuvan University
Khatmandu, Nepal

O. P. Malik
The University of Calgary
Calgary, Alberta, Canada

Abstract: A scheme for power factor correction of an induction machine by using an auxiliary stator winding in conjunction with a PWM voltage inverter and a capacitor on the dc side of the inverter is described in this paper. The operating principle of the proposed scheme is explained and a criterion is developed to select the optimum value of the power factor at which the scheme gives best performance. Simulation and experimental studies show that the harmonic components in the PWM output voltage can be reduced to get nearly sinusoidal current.

Keywords: Power factor correction; induction machines; PWM inverter.

INTRODUCTION

Induction machines normally draw lagging current with a fairly low power factor (pf). Poor pf adversely affects the economics of the transmission and distribution systems. Use of fixed shunt capacitors cannot fulfill the requirement of large induction machines with varying load. Solid state reactive power compensators are becoming more popular because they offer advantages over the conventional techniques [1].

Use of a solid state VAr compensator with a PWM inverter to compensate for reactive power at a utility load bus has been explained in the literature [2, 3], and the PWM technique has been used as a means of reducing the size of reactive components. The basic principle involved in this PWM technique has been used to develop a scheme for pf correction of an induction motor.

The proposed scheme uses an auxiliary winding in conjunction with a PWM voltage inverter and a capacitor on the dc side of the inverter. A mathematical model of the proposed scheme is developed. Results of simulation studies and some experimental tests described in the paper show that the harmonic components in the PWM output voltage can be reduced to get a nearly sinusoidal current waveform.

PROPOSED SCHEME

An induction motor having two sets of stator windings, wound in the same slots but electrically

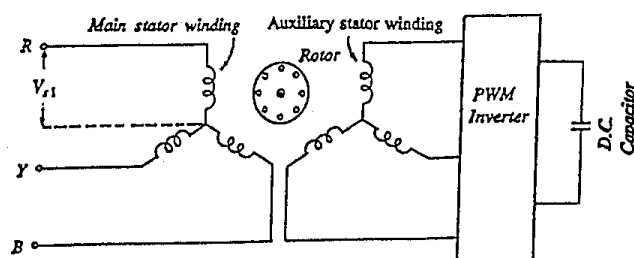


Fig. 1. Schematic diagram of the proposed scheme

isolated, is employed in the proposed scheme. One set, the main winding, is connected to the supply, while the other set, the auxiliary winding, is supplied by a PWM voltage inverter as shown in Fig. 1. The main idea is to control the magnitude and phase of the PWM inverter output voltage with respect to the main winding supply voltage so that the main winding will carry mainly the active power and the auxiliary winding will carry mainly the reactive power. Since the auxiliary winding carries mainly the reactive power, the dc voltage of the PWM voltage inverter can be supported by a dc capacitor and need not be connected to a power source [3].

Based on the equivalent circuit of a three winding transformer, the equivalent circuit of the induction machine having a double stator winding can be realized as shown in Fig. 2. The common mutual leakage reactance, X_{lm} , in Fig. 2 represents the fact that the two sets of stator windings occupy the same slots and are, therefore, mutually coupled by a component of leakage flux. The effect of the PWM voltage inverter is represented by an ac voltage source $V_{s2}(m)$ in the equivalent circuit. It will have fundamental as well as some harmonics.

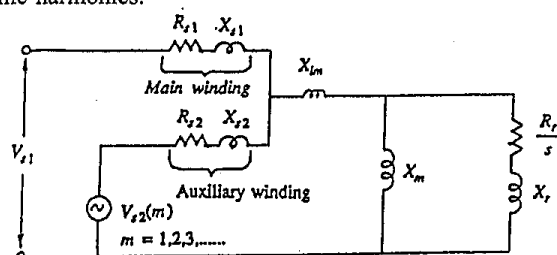


Fig. 2. Equivalent circuit of the proposed scheme

OPERATING PRINCIPLE

The operating principle of the proposed scheme can be explained with the help of the modified equivalent circuit shown in Fig. 3, in which the PWM inverter has been replaced by a fictitious variable capacitor having reactance X_{c2} . It is possible to make the imaginary part of the total impedance as seen from the main terminals equal to zero at some particular value of X_{c2} and thus

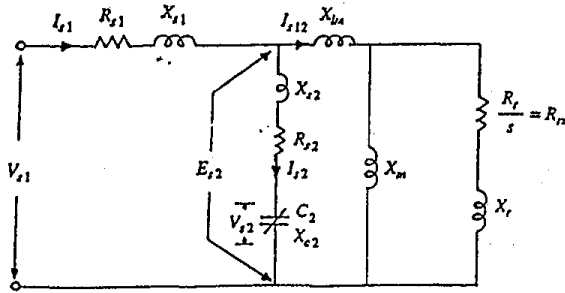


Fig. 3. Modified equivalent circuit

achieve unity pf .

For analysis, the equivalent circuit shown in Fig. 3 is simplified to that shown in Fig. 4 where,

$$X_{ceq} = X_{c2} - X_{s2} \quad (1)$$

$$R_{eq} = \text{Re} [j X_{lm} + (R_r/s + j X_r) // j X_m] \quad (2)$$

$$X_{eq} = \text{Im} [j X_{lm} + (R_r/s + j X_r) // j X_m] \quad (3)$$

The condition for unity pf is

$$\text{Im} [Z_1 + Z_2 // Z_3] = 0 \quad (4)$$

By mathematical manipulation of eqn. (4), the condition for unity pf can be written as

$$A (X_{ceq})^2 + B (X_{ceq}) + C = 0 \quad (5)$$

where,

$$A = X_{eq} + X_{s1} \quad (6)$$

$$B = R_{eq}^2 + X_{eq}^2 + 2 X_{s1} X_{eq}$$

$$C = X_{s1} [(R_{s2} + R_{eq})^2 + X_{eq}^2] + R_{s2}^2 X_{eq}$$

Hence,

$$X_{ceq} = (-B \pm \sqrt{B^2 - 4AC}) / 2A \quad (7)$$

provided $(B^2 - 4AC) \geq 0$.

Then,

$$X_{c2} = X_{ceq} + X_{s2} \quad (8)$$

The above analysis shows that unity pf can be obtained at two different values of X_{c2} for a particular value of slip. When the circuit is operated at one of these two values of X_{c2} , the fictitious capacitor will have some voltage V_{s2} across it with a phase angle θ with respect to the main supply voltage V_{s1} . If the PWM voltage inverter can be controlled to produce the same output voltage V_{s2}

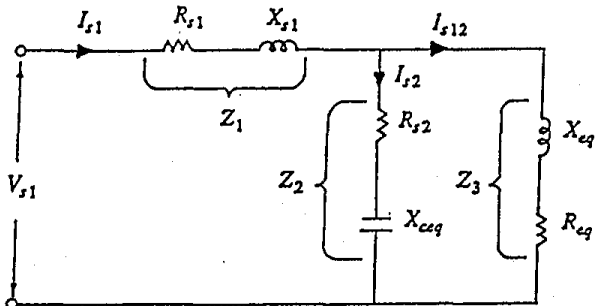


Fig. 4. Simplified equivalent circuit

with phase angle θ , unity pf can be obtained at the input

terminals. Hence, the effect of PWM inverter can be represented by a variable capacitor C_2 in the equivalent circuit as shown in Fig. 3.

A phasor diagram of the proposed scheme for unity pf is shown in Fig. 5. The function of the PWM inverter is to produce a voltage V_{s2} with an appropriate phase and angle θ so that the current through the auxiliary stator winding I_{s2} has a leading reactive component $I_{s2} \sin \theta$ of the phasor current I_{s12} drawn by the motor in the absence of the auxiliary stator winding.

The reactance X_{c2} is the ratio of the PWM output voltage V_{s2} and the current through the auxiliary winding I_{s2} , where V_{s2} is the fundamental component of the PWM output voltage. In practice, the PWM inverter output voltage has a fundamental as well as harmonic components. Performance under this condition is discussed later.

Variation of the imaginary component of the total impedance with X_{ceq} for different values of slip for a 90hp, 500V, 8 pole, 60Hz induction motor [4] is shown in Fig. 6. It can be seen from this figure that unity pf can be obtained at different values of slip (load). For a particular value of slip, pf becomes unity at two different values of X_{ceq} . The higher value of X_{ceq} corresponds to higher values of V_{s2} which results in a lower value of current through the auxiliary winding. To reduce loss in the auxiliary winding, PWM inverter should be controlled to obtain a higher value of X_{ceq} .

It is also seen from Fig. 6 that the value of X_{ceq} required to produce unity pf at higher load is comparatively smaller than that required at light load. Therefore at higher load, high current will flow through the auxiliary winding for unity pf operation, increasing losses in the auxiliary winding and lower efficiency of the motor. Hence, operating the machine at unity pf may not give the most desired performance.

OPTIMUM VALUE OF POWER FACTOR

Characteristics, such as I_{s2} versus pf , loss in R_{s2} versus pf and efficiency versus pf calculated at fixed values of slip for the 90hp induction machine mentioned above show that I_{s2} and power loss in R_{s2} are very high at unity pf , and they decrease rapidly within a small range of pf , such as 1.0 to 0.95 lag. After that there is only a small decrease in I_{s2} and power loss in R_{s2} with decrease in pf . Efficiency characteristics for two values of slip are given in Fig. 7. Figure 7(a) shows that the machine has

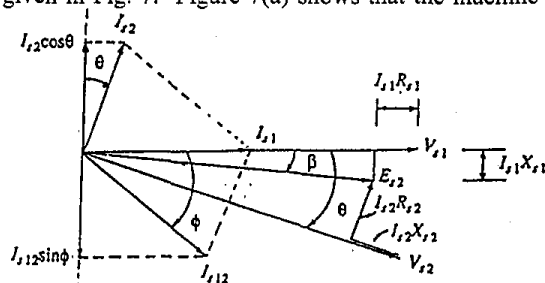


Fig. 5. Phasor diagram for unity power factor operation

maximum efficiency at a pf which lies between a and b ,

which region also happens to lie outside the region where I_{s2} and power loss in R_{s2} are high. Hence, the criterion for maximum efficiency can be used to select the optimum value of pf for a particular value of slip.

Comparing the results given in Figs. 7(a) and 7(b) it is seen that the pf at which the maximum efficiency occurs varies with load. Using the criterion of maximum efficiency to select the power factor means that the machine will operate at a fairly low power factor at light loads. To avoid operation at low power factor, it is better to put a lower limit on the power factor, e.g. 0.85, for the operation of the machine. Because of the narrow range over which the efficiency of the machine varies with power factor (Fig. 7(b)), the machine will still operate close to the maximum efficiency.

CONTROL STRATEGY

The PWM voltage inverter has to produce a particular value of V_{s2} (1) with a particular value of phase angle θ in order to maintain the desired pf . Hence, at a particular load, the required magnitude and phase of V_{s2} (1) has to be obtained and kept constant with the help of a controller.

Referring to the phasor diagram, Fig. 5, active power associated with the auxiliary stator winding is given by:

$$P_{s2} = E_{s2} I_{s2} \sin(\theta - \beta) \text{ W/phase} \quad (9)$$

The auxiliary winding receives this power from the main stator winding by transformer action. Current through the auxiliary winding is given by

$$\bar{I}_{s2} = [\bar{E}_{s2} - \bar{V}_{s2}(1)] / (R_{s2} + jX_{s2}) \quad (10)$$

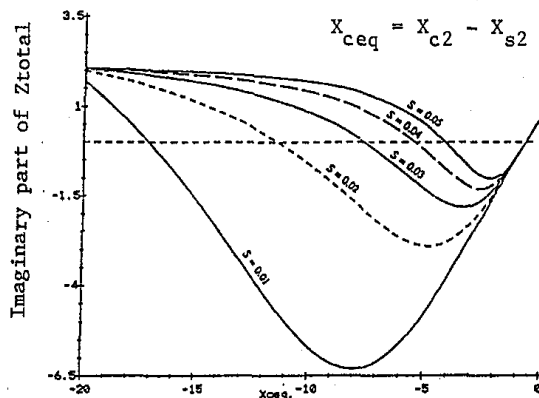


Fig. 6. Imaginary part of total impedance versus X_{ceq}

Control strategy can be realized with the help of eqns. (9) and (10). Reactive power in the main stator winding can be sensed and compared with the desired reactive power, Q_{ref} , and the error signal thus obtained can be used to operate the controller. The controller selects proper modulation index for the PWM inverter to produce the required magnitude of V_{s2} (1) to give the desired power factor.

Another variable that has to be controlled is the phase angle θ related to the active power in the auxiliary

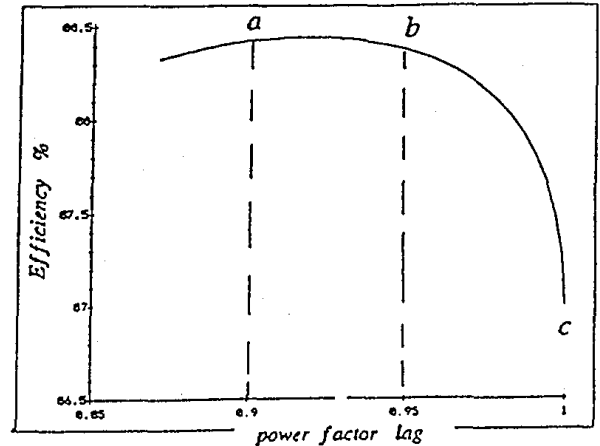
winding. If

$$E_{s2} I_{s2} \sin(\theta - \beta) = I_{s2}^2 R_{s2} \quad (11)$$

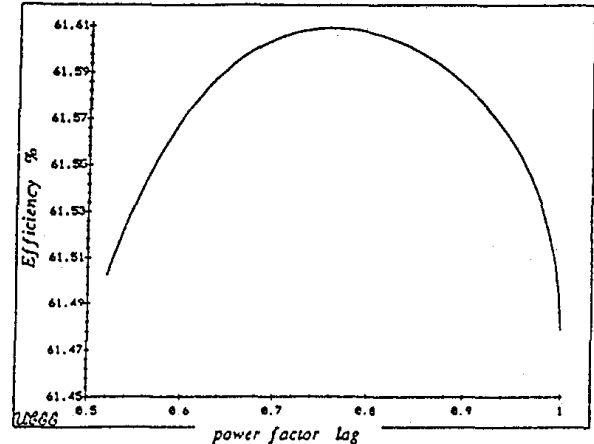
$$= P_{loss_{aux}} \text{ W / phase}$$

then there will be no power flow into the dc capacitor and the dc voltage across the capacitor will remain constant.

The dc voltage across the capacitor can be used to regulate the phase angle θ . By comparing this voltage with the $V_{dc}(ref)$, the error signal can be used to operate the controller to give the desired value of θ . A block diagram for the control scheme is shown in Fig. 8.



(a) 0.04 pu slip



(b) 0.0015 pu slip

Fig. 7. Induction motor characteristics

EFFECT OF HARMONICS

The PWM output voltage has a fundamental and harmonic components. Fundamental currents through different branches of the circuit can be calculated from the equivalent circuit shown in Fig. 3 by using only the fundamental component of the inverter output voltage V_{s2} (1). The main supply voltage V_{s1} is assumed to have only the fundamental component.

Harmonic currents through different branches of the circuit can be also calculated by using the equivalent

circuit of Fig. 3 with its input terminals short-circuited. Harmonic current of each order is calculated one at a time with the reactances in the equivalent circuit suitably modified corresponding to the order of harmonic. Total current through different branches of the circuit is then given as a sum of the individual harmonic currents through the branch.

The voltage harmonics in the PWM output voltage V_{s2} can be reduced by operating the inverter at a higher carrier frequency. The machine inductances also provide substantial filtering action to reduce harmonics in the current as they offer higher reactance to the voltage of the higher order harmonics. Hence, if the PWM inverter is operated at a sufficiently high carrier frequency, the distortion factor in the main input current I_{s1} can be reduced to an acceptable value without any additional filter.

DC CAPACITOR RATING

The controller described above maintains across the capacitor constant average value of dc voltage required for inverter operation. However, the dc voltage will have some ripple which will result in a higher distortion factor in the main supply input current I_{s1} . The rating of the capacitor has to be selected such that the ripple factor for the dc voltage is less than 5 percent [5].

The ripple factor, K_v , is given by:

$$K_v = \sqrt{\sum_{m=0}^{\infty} V_c^2(m)} / V_{dc} \quad (12)$$

where $V_c(m)$, the rms value of mth. voltage harmonic across the dc capacitor, is

$$V_c(m) = I_{dc}(m) X_c / m \quad (13)$$

V_{dc} is average value of dc voltage across the capacitor,

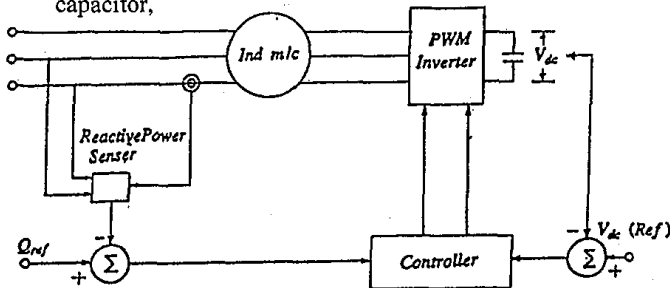


Fig. 8. Control block diagram for the proposed scheme

$I_{dc}(m)$ is rms value of mth. current harmonic through the dc capacitor.

X_c , reactance of the capacitor at fundamental frequency, is

$$X_c = K_v \cdot V_{dc} / \sqrt{\sum_{m=6,12,\dots}^{\infty} I_{dc}^2(m) / m^2} \quad (14)$$

Fourier analysis of the current waveform shows that the 12th, 18th, . . . harmonic components are extremely

small. Thus the expression X_c can be simplified by considering only the 6th harmonic component of the current through the dc capacitor.

Hence,

$$X_c = 6 k_v V_{dc} / I_{dc}(6) = 1 / \omega C \quad (15)$$

kVAR rating of the required capacitor is

$$kVAR = V_{dc} I_{dc}(rms) \quad (16)$$

SIMULATION STUDIES

A computer program has been written to simulate the various elements of the proposed scheme described above, and to study the operating conditions and performance. The induction machine can be analyzed with or without auxiliary winding. With the auxiliary winding, it is possible to study operation at unity pf or calculate the required magnitude and phase of PWM inverter output voltage to give optimum power factor. Newton-Raphson method is used to determine the switching pattern of PWM inverter waveform required to produce the desired output voltage, V_{s2} . Fourier components of V_{s2} , performance in the presence of harmonics in V_{s2} and the rating of various components also can be calculated.

Studies on 90 hp Machine

The 90 hp machine, mentioned above, operating at light load ($s = 0.0015$ pu) without an auxiliary winding draws a current of 14.38A at 0.224 pf lag. The output power is 1.6 hp with an efficiency of 43.22%. With the power factor corrected by using the auxiliary winding in conjunction with the PWM inverter operated with a frequency ratio (FR) of 24, the optimum value of pf is calculated to be unity, and the required PWM inverter output voltage is 276.72V with a phase angle of 0.75 deg lag with reference to V_{s1} . Harmonic current components are very significant as shown by the distortion factors in Table I.

The distortion factors of currents for FR of 42 and 54 are also shown in Table I. It can be seen that the distortion factors are reduced as the FR is increased. Waveforms of I_{s1} and I_{s2} reconstructed from the first 48 Fourier components of the currents are shown in Figs. 9 and 10 for FR of 42 and 54 respectively. In Fig. 9, the fundamental component of I_{s1} is exactly in phase with V_{s1} and the fundamental component of I_{s2} leads V_{s1} by an angle of 89.2 deg.

Results at full load ($s = 0.055$ pu) are summarized in Tables II and III. It can be seen from Table III that operating the scheme at unity pf does not give the best performance. Current waveforms for one case, i.e. FR of 54 are shown in Fig. 11. Based on results in Table II, Fig. 11 and the waveforms for FR of 24 and 42 (not shown here) it is seen that the current waveforms are comparatively less distorted than at light load conditions. At optimum power factor, rating of the dc capacitor is 1000V, 60 μ F, 40 kVAR or 2 units each of 500V, 120 μ F, 20 kVAR.

Table I
Distortion Factors for $s = 0.0015$ pu

Variable	FR = 24	FR = 42	FR = 54
I_{s1}	2.3	0.1194	0.0745
I_{s2}	0.557	0.0285	0.0185
I_r	1.66	0.0237	0.012

Table II
Distortion Factors for $s = 0.055$ pu

Variable	FR = 24	FR = 42	FR = 54
I_{s1}	0.05160	0.00634	0.00328
I_{s2}	0.24560	0.00474	0.00170
I_r	0.02940	0.01990	0.00172

Table III
Performance at two pf s

	Unity pf	Optimum pf
I_{s1} (1) A	98.73	89.65
I_{s2} (1) A	77.70	29.20
I_r (1) A	105.78	84.21
V_{s2} (1) V	328.56	276.85
Phase of V_{s2} deg	-14.15	-10.60
Loss in R_{s2} W	2354	333
Inverter rating kVA	76.50	24.26
Input power W	85505	70527
Output power W	73485	62795
Efficiency %	85.94	89.04

Studies on 2hp Machine

A commercial 2hp, 200V, 4 pole, 60Hz induction machine was rewound with one main and one auxiliary winding for the purpose of testing it in the laboratory. Its parameters were obtained by test and used in the simulation studies.

Distortion factors for slips of 0.0066pu and 0.06pu are given in Table IV. Waveform of I_{s1} for these two slips with FR of 40 is shown in Fig. 12. This figure, and the results for FR of 20 and 54 (not included here) show that the waveforms become less distorted both with higher FRs and higher loads.

Table IV
Distortion factors for 2hp machine

Variable	FR = 20	FR = 40	FR = 54
$s = 0.0066$ pu			
I_{s1}	2.0890	0.0864	0.0536
I_{s2}	0.4280	0.0169	0.0105
I_r	2.1710	0.0284	0.0140
$s = 0.06$ pu			
I_{s1}	0.1818	0.0054	0.0052
I_{s2}	0.5172	0.0222	0.0115
I_r	0.1174	0.0031	0.0016

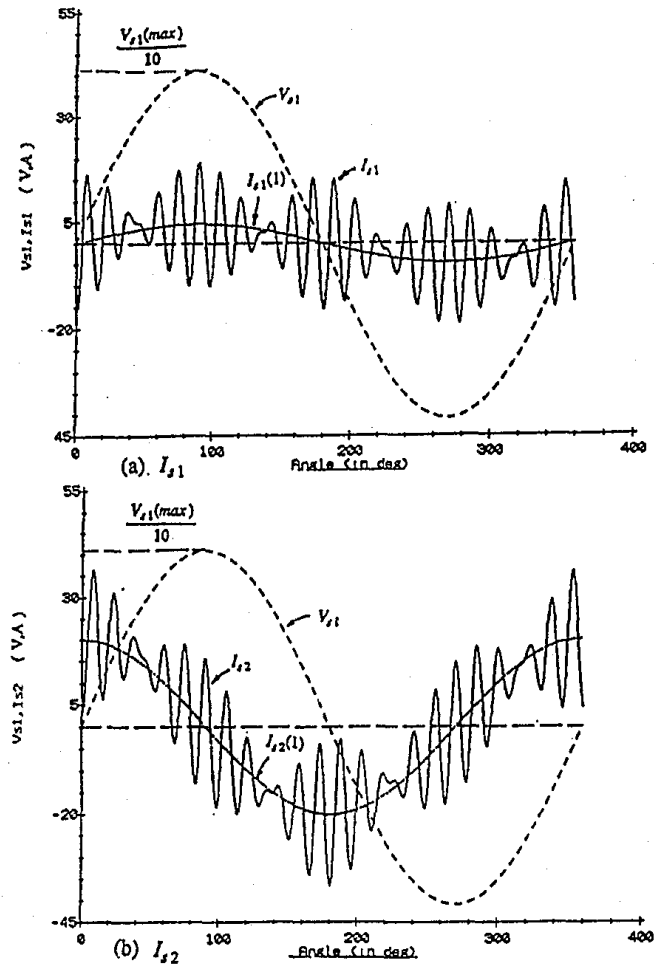


Fig. 9. Waveforms at 0.0015 pu slip, FR = 24 for 90hp machine deg. giving a pf of 0.9 lag

EXPERIMENTAL STUDIES

Experimental tests were conducted on the 2hp induction machine to test the basic principle of the proposed power factor correction scheme. The main winding was connected to the 3-phase ac supply. Due to the lack of proper equipment and control system, implementation was not exactly as used in the simulation studies. In the first phase, the experiment was set up by replacing the PWM voltage inverter by sinusoidal ac voltage source (an auto-transformer). In the next phase, the auxiliary winding was supplied by a PWM inverter with the dc side supplied by an independently controlled dc voltage source. In this case, the inverter output voltage is exactly in phase with E_{s2} rather than adjustable lagging phase angle.

With the induction motor running at light load, $s = 0.0066$ pu, coupled to the shaft of a dc machine, the applied voltage, input current, input power and speed were measured with the auxiliary winding supplied by an autotransformer. The supply from the autotransformer was connected to the auxiliary winding when the emf E_{s2} and V_{s2} were exactly equal and in phase. Then the magnitude of V_{s2} was adjusted to correct the power factor. The test was repeated with the auxiliary winding

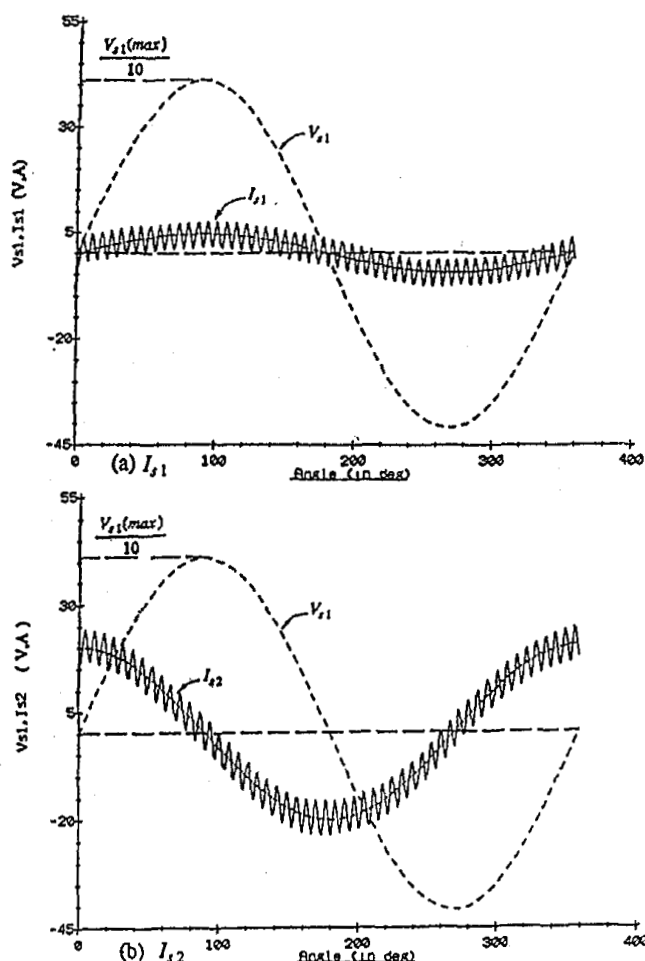


Fig. 10. Waveforms for 90hp machine at 0.0015 pu slip, FR = 54

supplied by the PWM inverter. Because the PWM inverter available in the laboratory can produce a maximum line to line voltage of 100V the induction machine was operated at 95V for these tests.

The results of these tests are summarized in Table V and the waveforms of I_s are given in Fig. 13. The input current with FR of 20, Fig. 13(b), is highly distorted and its value of 4.6A, Table V, includes the fundamental and harmonic components. Phase angle between the fundamental components of V_s and I_s as measured was 32.5 deg which gives a pf of 0.84 lag.

Table V
Experimental Results

FR	I_s A	P_{in} W	pf (calc.)	A_{s2} A
Autotransformer	1.4	180	0.78	1.0
PWM Inverter				
20	4.6	225	0.34	5.2
40	2.5	195	0.47	3.2

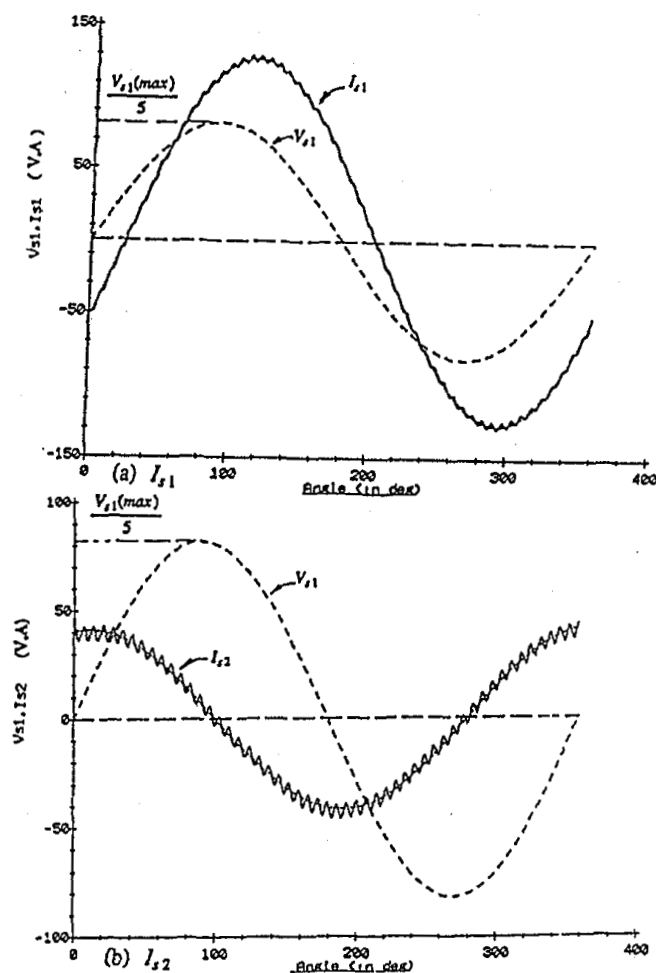


Fig. 11. Waveforms for 90hp machine at 0.055 pu slip, FR = 40

With FR of 40, the distortion in I_s is significantly reduced as shown in Fig. 13(c). Phase angle between the fundamental components of V_s and I_s in this case was measured to be 24 deg, giving a pf of 0.9 lag.

Even though only the basic principle of the proposed scheme for pf correction of an induction machine has been tested experimentally, the results do show trends similar to those obtained in simulation studies.

CONCLUSIONS

A scheme for pf correction of an induction machine by using an auxiliary stator winding in conjunction with a PWM voltage inverter and a single capacitor on the dc side of the inverter is proposed. It can provide continuous control of reactive power and can be an economical alternative to the fixed ac capacitors with thyristor controlled reactor schemes. By operating the PWM inverter at higher chopping frequencies, harmonics can be significantly reduced.

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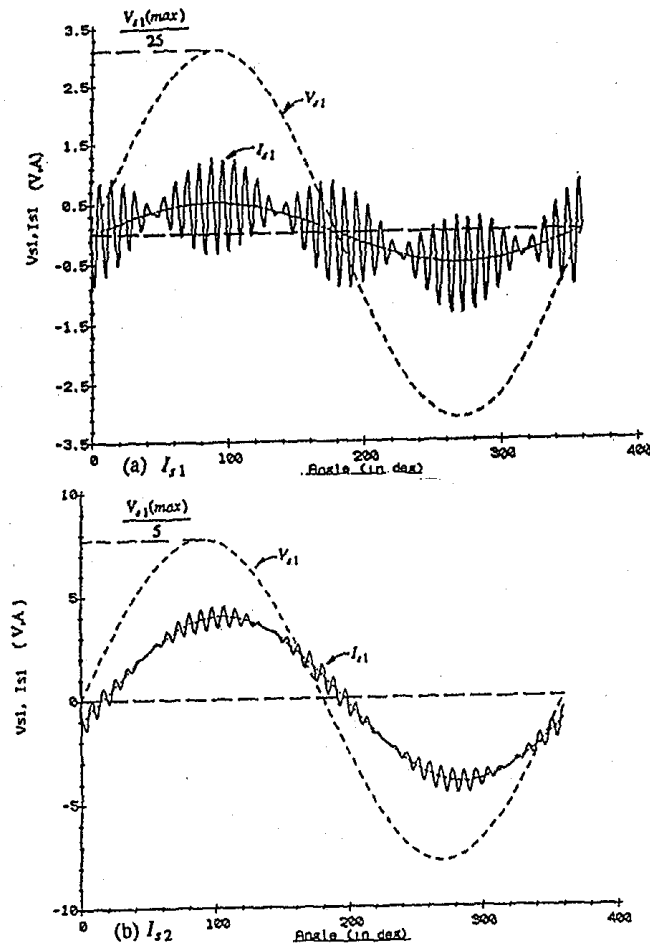


Fig. 12. Waveforms for 2hp machine at 0.06 pu slip, FR = 40

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BIOGRAPHY

Tamrakar, Indraman obtained B.Sc. in Electrical Engineering from Punjab University, India in 1983 and an M.Sc. Degree in Electrical Engineering from the University of Calgary, Canada in 1992. He has been teaching at Institute of Engineering, Tribhuvan University, Nepal since 1983 and is at present the Head of Electrical Engineering Department there.

Malik, O.P. (M'66-SM'69-F;87) graduated in Electrical Engineering in 1952, obtained M.E. degree in 1962 and the Ph.D. degree and D.I.C., London in 1965. From 1952-1961 he worked with electrical utilities in India. He has been teaching at the University of Calgary since 1968, where he is at present a Faculty Professor.

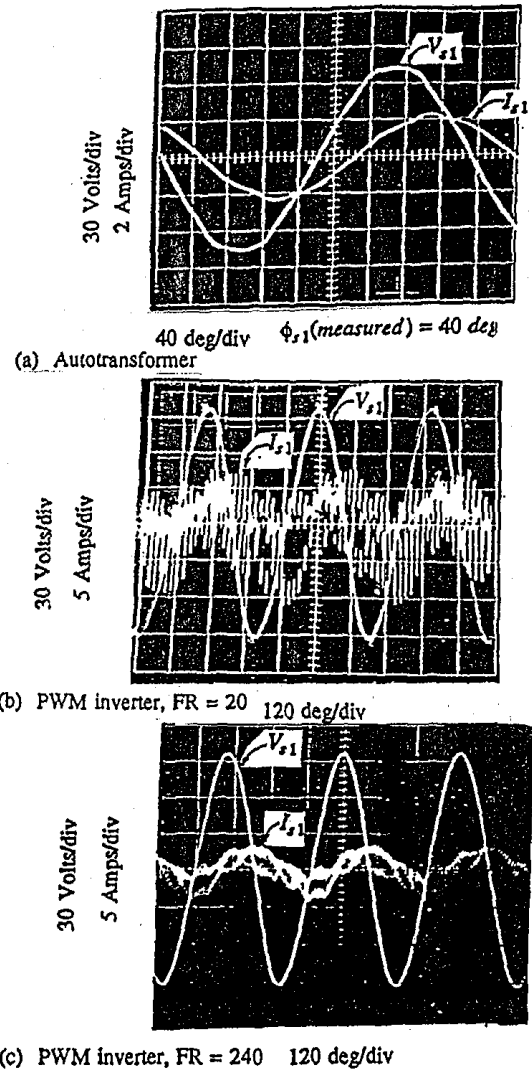


Fig. 13. Waveforms of V_{s1} and I_{s1} for 2hp machine at 0.0066 pu slip